

第一题:

$$1-2. \min z = -3x_1 + 4x_2 - 2x_3 + 5x_4.$$

$$s.t. \begin{cases} 4x_1 - x_2 + x_3 - x_4 = -2 \\ x_1 + x_2 + x_3 - x_4 \leq 14 \\ -2x_1 + 3x_2 - x_3 + 2x_4 \geq 2 \\ x_1 \geq 0, x_3 \geq 0, x_2 \leq 0, x_4 \text{ 无限制} \end{cases}$$

解: 首先在第一个约束条件两端乘以-1, 使得约束条件右端项常数均为非负数。

令 $x_2' = -x_2$, $x_4 = x_4' - x_4''$, $x_4' \geq 0, x_4'' \geq 0$, 引入松弛变量 $x_5 \geq 0, x_6 \geq 0$

则原问题的标准形式为:

$$\max (-z) = 3x_1 + 4x_2' + 2x_3 - 5x_4' + 5x_4''$$

$$s.t. \begin{cases} -4x_1 - x_2' - x_3 + x_4' - x_4'' = 2 \\ x_1 - x_2' + x_3 - x_4' + x_4'' + x_5 = 14 \\ -2x_1 - 3x_2' - x_3 + 2x_4' - 2x_4'' - x_6 = 2 \\ x_1, x_2', x_3, x_4', x_4'', x_5, x_6 \geq 0 \end{cases}$$

其矩阵形式为

$$\max z = C^T X$$

$$s.t. \begin{cases} AX = b \\ X \geq 0 \end{cases}$$

其中

$$A = \begin{pmatrix} -4 & -1 & -1 & 1 & -1 & 0 & 0 \\ 1 & -1 & 1 & -1 & 1 & 1 & 0 \\ -2 & -3 & -1 & 2 & -2 & 0 & -1 \end{pmatrix}, C = (3, 4, 2, -5, 5, 0, 0)^T,$$

$$b = (2, 14, 2)^T, X = (x_1, x_2', x_3, x_4', x_4'', x_5, x_6)^T$$

$$1-4. \max z = 2x - 3y$$

$$s.t. \begin{cases} |x| + |y| = 3 \\ x - y = 1 \end{cases}$$

解: 令 $x_1 = \frac{|x| - x}{2}$, $x_2 = \frac{|x| + x}{2}$, $x_3 = \frac{|y| - y}{2}$, $x_4 = \frac{|y| + y}{2}$

则易得 $x_i \geq 0$ ($i=1, 2, 3, 4$). 且有 $|x| = x_1 + x_2$, $x = x_2 - x_1$, $|y| = x_3 + x_4$, $y = x_4 - x_3$.

原问题的标准形式为:

$$\max z = -2x_1 + 2x_2 + 3x_3 - 3x_4$$

$$s.t. \begin{cases} x_1 + x_2 + x_3 + x_4 = 3 \\ -x_1 + x_2 + x_3 - x_4 = 1 \\ x_i \geq 0 \quad (i=1, 2, 3, 4) \end{cases}$$

其矩阵形式为:

$$\max z = C^T X$$

$$s.t. \begin{cases} AX = b \\ X \geq 0 \end{cases}$$

其中: $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \end{pmatrix}, C = (-2, 2, 3, -3)^T$

$$b = (3, 1)^T, X = (x_1, x_2, x_3, x_4)^T$$

第二题:

2-1. 解: 本题中线性规划问题已经是标准形式, 矩阵表达为:

$$\begin{aligned} \max \quad & z = C^T X \\ \text{s.t.} \quad & \begin{cases} AX = b \\ X \geq 0 \end{cases} \end{aligned}$$

其中: $A = \begin{pmatrix} 12 & 3 & 6 & 3 & 0 & 0 \\ 8 & 1 & -4 & 0 & 2 & 0 \\ 3 & 0 & 0 & 0 & 0 & -1 \end{pmatrix}$, $C = (3, 1, 2, 0, 0, 0)^T$

$b = (9, 10, 0)^T$, $X = (x_1, x_2, x_3, x_4, x_5, x_6)^T$

记A的6个列向量按顺序为 $(p_1, p_2, p_3, p_4, p_5, p_6)$, 从中选出3个列向量共可组成 $C_6^3 = 20$ 个子矩阵.

(1) $B = (p_1, p_2, p_3) = \begin{pmatrix} 12 & 3 & 6 \\ 8 & 1 & -4 \\ 3 & 0 & 0 \end{pmatrix}$

$|B| \neq 0$ $B^{-1}b = (0, \frac{16}{3}, -\frac{7}{6})^T$

$X^* = (0, \frac{16}{3}, -\frac{7}{6}, 0, 0, 0)^T$ 为基解.

(2) $B = (p_1, p_2, p_4)$, $|B| \neq 0$

$B^{-1}b = (0, 10, -7)^T$

$X^* = (0, 10, 0, -7, 0, 0)^T$ 为基解

(3) $B = (p_1, p_2, p_5)$, $|B| \neq 0$

$B^{-1}b = (0, 3, \frac{7}{2})^T$

$X^* = (0, 3, 0, 0, \frac{7}{2}, 0)^T$ 为基可行解.

$z^* = 3$.

(4) $B = (p_1, p_2, p_6)$, $|B| \neq 0$

$B^{-1}b = (\frac{7}{4}, -4, \frac{3}{4})^T$

$X^* = (\frac{7}{4}, -4, 0, 0, 0, \frac{3}{4})^T$ 为基解.

(5) $B = (p_1, p_3, p_4)$, $|B| \neq 0$

$B^{-1}b = (0, -\frac{5}{2}, 8)^T$

$X^* = (0, 0, -\frac{5}{2}, 8, 0, 0)^T$ 为基解.

(6) $B = (p_1, p_3, p_5)$, $|B| \neq 0$

$B^{-1}b = (0, \frac{3}{2}, 8)^T$

$X^* = (0, 0, \frac{3}{2}, 0, 8, 0)^T$ 为基可行解.

$z^* = 3$

(7) $B = (p_1, p_3, p_6)$, $|B| \neq 0$

$B^{-1}b = (1, -\frac{1}{2}, 3)^T$

$X^* = (1, 0, -\frac{1}{2}, 0, 0, 3)^T$ 为基解.

(8) $B = (p_1, p_4, p_5)$, $|B| \neq 0$

$B^{-1}b = (0, 3, 5)^T$

$X^* = (0, 0, 0, 3, 5, 0)^T$ 为基可行解.

$z^* = 0$.

(9) $B = (p_1, p_4, p_6)$, $|B| \neq 0$

$B^{-1}b = (\frac{15}{4}, -2, \frac{15}{4})^T$

$X^* = (\frac{15}{4}, 0, 0, -2, 0, \frac{15}{4})^T$ 为基解.

(10) $B = (p_1, p_5, p_6)$, $|B| \neq 0$

$B^{-1}b = (\frac{3}{4}, 2, \frac{9}{4})^T$

$X^* = (\frac{3}{4}, 0, 0, 0, 2, \frac{9}{4})^T$ 为基可行解.

$z^* = \frac{9}{4}$

(11) $B = (p_2, p_3, p_6)$, $|B| \neq 0$

$B^{-1}b = (\frac{16}{3}, -\frac{7}{6}, 0)^T$

$X^* = (0, \frac{16}{3}, -\frac{7}{6}, 0, 0, 0)^T$ 为基解.

(12) $B = (p_2, p_4, p_6)$, $|B| \neq 0$

$B^{-1}b = (10, -7, 0)^T$

$X^* = (0, 10, 0, -7, 0, 0)^T$ 为基解.

(13) $B = (p_2, p_5, p_6)$, $|B| \neq 0$

$B^{-1}b = (3, \frac{7}{2}, 0)^T$

$X^* = (0, 3, 0, 0, \frac{7}{2}, 0)^T$ 为基可行解.

$z^* = 3$.

(14) $B = (p_3, p_4, p_6)$, $|B| \neq 0$

$B^{-1}b = (-\frac{5}{2}, 8, 0)^T$

$X^* = (0, 0, -\frac{5}{2}, 8, 0, 0)^T$ 为基解.

$$(15) B = (p_3, p_5, p_6), |B| \neq 0$$

$$B^{-1}b = (\frac{3}{2}, 8, 0)^T$$

$$X^* = (0, 0, \frac{3}{2}, 0, 8, 0)^T \text{ 为基可行解}$$

$$z^* = 3$$

$$\text{又当 } B = (p_2, p_3, p_4), |B| = 0$$

$$B = (p_2, p_3, p_5), |B| = 0$$

$$B = (p_2, p_4, p_5), |B| = 0$$

$$B = (p_3, p_4, p_5), |B| = 0 \text{ 均非基解}$$

$$\text{综上所述, 最优解为: } (0, 0, \frac{3}{2}, 0, 8, 0)^T, (0, 3, 0, 0, \frac{7}{2}, 0)^T$$

$$(16) B = (p_4, p_5, p_6), |B| \neq 0$$

$$B^{-1}b = (3, 5, 0)^T$$

$$X^* = (0, 0, 0, 3, 5, 0)^T \text{ 为基可行解}$$

$$z^* = 0$$

2-2. 解: 本题中线性规划标准形式的矩阵表达式为:

$$\begin{aligned} \max \quad & z = C^T X \\ \text{s.t.} \quad & \begin{cases} AX = b \\ X \geq 0 \end{cases} \end{aligned}$$

$$\text{其中, } A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 2 & 1 & 2 \end{pmatrix}, C = (-5, 2, -3, -2)^T$$

$$b = (7, 3)^T, X = (x_1, x_2, x_3, x_4)^T$$

记A的列向量按顺序为 (p_1, p_2, p_3, p_4) , 从中选出2个列向量并可组成 $C_4^2 = 6$ 个子矩阵.

$$(1) B = (p_1, p_2), |B| \neq 0$$

$$B^{-1}b = (-4, \frac{11}{2})^T$$

$$X^* = (-4, \frac{11}{2}, 0, 0)^T \text{ 为基解}$$

$$(2) B = (p_1, p_3), |B| \neq 0$$

$$B^{-1}b = (\frac{7}{2}, \frac{11}{5})^T$$

$$X^* = (\frac{7}{2}, 0, \frac{11}{5}, 0)^T \text{ 为基可行解}$$

$$z^* = -\frac{43}{5}$$

$$(3) B = (p_1, p_4), |B| \neq 0$$

$$B^{-1}b = (-\frac{1}{3}, \frac{11}{6})^T$$

$$X^* = (-\frac{1}{3}, 0, 0, \frac{11}{6})^T \text{ 为基解}$$

$$(4) B = (p_2, p_3), |B| \neq 0$$

$$B^{-1}b = (\frac{1}{2}, 2)^T$$

$$X^* = (0, \frac{1}{2}, 2, 0)^T \text{ 为基可行解}$$

$$z^* = -5$$

$$(5) B = (p_2, p_4), |B| \neq 0$$

$$B^{-1}b = (-\frac{1}{2}, 2)^T$$

$$X^* = (0, -\frac{1}{2}, 0, 2)^T \text{ 为基解}$$

$$(6) B = (p_3, p_4), |B| \neq 0$$

$$B^{-1}b = (1, 1)^T$$

$$X^* = (0, 0, 1, 1)^T \text{ 为基可行解}$$

$$z^* = -5$$

$$\text{最优解为: } (0, \frac{1}{2}, 2, 0)^T, (0, 0, 1, 1)^T$$

第三题:

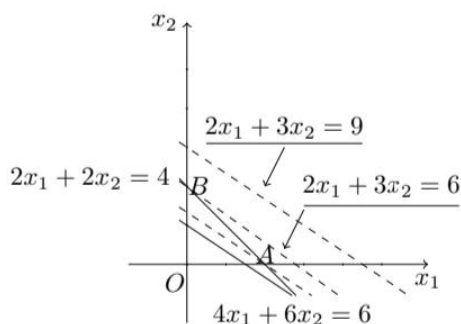


图 1.1 图解法求解3(1)

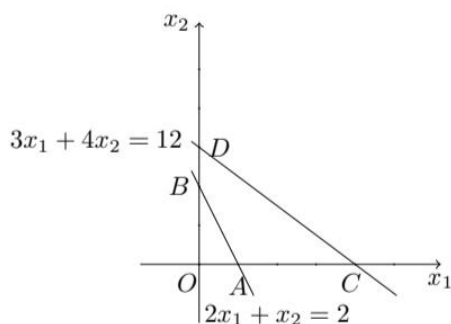


图 1.2 图解法求解3(2)

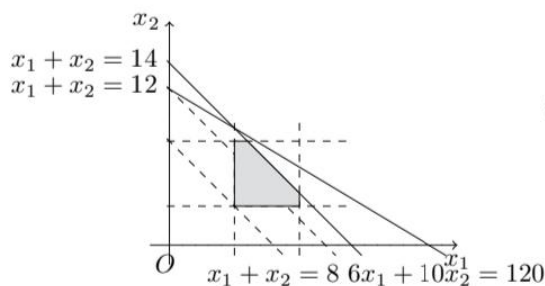


图 1.3 图解法求解3(3)

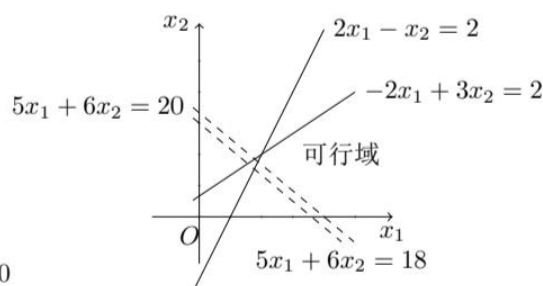


图 1.4 图解法求解3(4)

解 (1)参见图(1.1)，可行域为线段 AB 在第一象限的右上区域。虚线为目标函数，可见从右上往左下移动时目标函数值减少。唯一最优解为直线 $2x_1 + 2x_2 = 4$ (即 $x_1 + x_2 = 2$)与 x_1 轴的交点： $x^* = (2, 0)^T$ ，最优值为 $x^* = 4$ 。

(2)参见图(1.2)，可行域为线段 CD 在第一象限的右上区域与三角形区域 ABO 的交集，显然为空集。即该问题无可行解。

(3)参见图(1.3)，可行域为图中的阴影部分。目标函数与直线 $x_1 + x_2 = 14$ 重合时取得极大值，故该问题有无穷多最优解，连接 $(6, 8)$ 、 $(10, 4)$ 两点线段中的所有点均为最优解。故最优解可表为 $x^* = \lambda(6, 8) + (1 - \lambda)(10, 4) = (10 - 4\lambda, 4 + 4\lambda)$, $0 \leq \lambda \leq 1$ 。

(4)参加图(1.4)，可行域为图中表示的开阔的区域，从画出的两条目标函数直线（虚线） $5x_1 + 6x_2 = 18$ 和 $5x_1 + 6x_2 = 20$ 可以看出该问题为无界解。

第四题：

解 (1) 参见图(1.5)，最优解为直线 $3x_1 + 4x_2 = 9$ 与 $5x_1 + 2x_2 = 8$ 的交点： $x^* = (1, 1.5)^T$ ，最优值为 $x^* = 17.5$ 。其单纯形表求解过程见表(1.1)。表(1.1)中第一次迭代相当于图(1.5) 中点 O ，第二次迭代对应点 A ，第三次迭代对应点 B 。

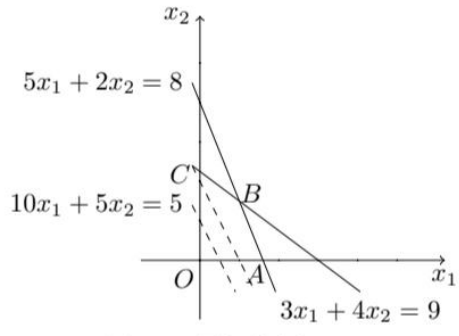


图 1.5 图解法求解4(1)

表 1.1 习题4(1)的初始单纯形表及其迭代

c_j			10	5	0	0	θ
c_B	X_B	b	x_1	x_2	x_3	x_4	
0	x_3	9	3	4	1	0	3
0	x_4	8	[5]	2	0	1	8/5
$\sigma_j = c_j - \sum_{i=1}^m c_i a_{ij}$			10	5	0	0	
0	x_3	21/5	0	[14/5]	1	-3/5	3/2
10	x_1	8/5	1	2/5	0	1/5	4
$\sigma_j = c_j - \sum_{i=1}^m c_i a_{ij}$			0	1	0	-2	
5	x_2	3/2	0	1	5/14	-3/14	
10	x_1	1	1	0	-1/7	2/7	
$\sigma_j = c_j - \sum_{i=1}^m c_i a_{ij}$			0	0	-5/14	-25/14	